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Inventor(s)

Perez, III; Armando et al.

NUT PLATE ASSEMBLY INCLUDING FLOATING NUT WITH LOCKING ARRANGEMENT

Abstract

A nut plate assembly includes a plate comprising an aperture defined therethrough, and a shell attached to the plate and extending along a center axis. The nut plate also includes a nut disposed within the shell. The nut is tiltable relative to the center axis when the nut is retained within the shell. A fastener is received within the nut plate assembly and engages the nut when the fastener is in a locked orientation. The nut includes an engagement surface that engages the fastener when the fastener is in the locked orientation.

Inventors: Perez, III; Armando (Los Angeles, CA), Komsitsky; Igor (Los Angeles, CA)

Applicant: Enduralock, LLC (Lenexa, KS)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Patent Application No. 63/525,558, filed on Jul. 7, 2023, the entire content of which is incorporated by reference.

BACKGROUND

[0003] The field of the disclosure relates generally to nut plates, and more specifically to nut plate assemblies including a floating nut that provides a quarter-turn locking arrangement with a fastener.

[0004] At least some known floating nut plate assemblies include a base or bottom plate with an opening. The nut plate assemblies support a nut or similar fastener element that is captured on the base by a retainer or cage component. Such nut plates facilitate holding components, for example panels, together when a fastener is engaged with the nut and tightened, while still enabling some movement between the components. In such known nut plates, the nut receives a fastener, such as a bolt, screw, or other threaded component element passed through an opening in the components and threaded into the nut. The opening in the component may be sized to enable the nut and fastener to move laterally to accommodate some movement between the fastened components. However, the floating nut plates may not provide sufficient tolerances or account for misalignment of the fastener and the floating nut plate. In addition, the fastener may need to be screwed into the nut to provide a desired level of tightness. However, it may be difficult to rotate the fastener the required amount and/or determine if the fastener is sufficiently tightened.

[0005] At least some known floating nut plates may be used in the aerospace industry. In aerospace applications, the types and numbers of fasteners for a panel assembly can be significant. Some panel fasteners for a particular panel assembly may have different lengths, while otherwise looking similar to other panel fasteners. When the panel assembly is removed, a user may typically place all the panel fasteners in a separate location to keep from misplacing the fasteners. However, when replacing the panel assembly, the user may inadvertently use an incorrect length fastener for a particular panel fastener location. This can lead to an improperly attached panel assembly.

[0006] Accordingly, there is a need for a nut plate assembly that provides a secure connection between two panels and is simple to install. In addition, the nut plate assembly should accommodate misalignment of the fastener and the nut.

BRIEF DESCRIPTION

[0007] In one aspect, a nut plate assembly includes a plate comprising an aperture defined therethrough, a shell attached to the plate and extending along a center axis, and a nut retained within the shell and comprising an inner surface and an engagement surface that extends at an angle relative to the inner surface. The inner surface defines an inner bore and has a groove extending along the inner bore and parallel to the center axis. The nut plate assembly is configured to receive a fastener extending through the aperture of the plate and into the inner bore in the nut. In an unlocked orientation, the fastener is configured to move through the groove. In a locked orientation, the engagement surface engages the fastener and prevents the fastener moving axially relative to the floating nut.

[0008] In another aspect, a fastener assembly includes a plate comprising an aperture defined therethrough, a shell attached to the plate and extending along a center axis, a nut retained within the shell and comprising an inner bore, a fastener extending through the plate and into the shell to engage the nut, and a bias member disposed within the shell and engaged with the nut. The fastener is rotatable between an unlocked orientation and a locked orientation when the fastener is engaged with the nut. The nut includes an engagement surface that engages the fastener when the fastener is in the locked orientation. The unlocked orientation and the locked orientation are 90° apart.

[0009] In yet another aspect, a method of assembling a nut plate assembly includes positioning a

nut within a shell extending along a center axis. The nut comprises an inner surface and an engagement surface that extends at an angle relative to the inner surface. The inner surface defines an inner bore and has a groove extending along the inner bore and parallel to the center axis. The method also includes coupling a plate to the shell to capture the nut within the shell. The plate comprises an aperture arranged to receive a fastener such that the fastener extends through the plate and into the shell to engage the nut. In an unlocked orientation, the fastener is configured to move through the groove. In a locked orientation, the engagement surface engages the fastener and prevents the fastener moving axially relative to the floating nut.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0011] FIG. 1 is a perspective view of a nut plate assembly;

[0012] FIG. 2 is another perspective view of the nut plate assembly;

[0013] FIG. 3 is a side view of the nut plate assembly;

[0014] FIG. 4 is a bottom view of the nut plate assembly;

[0015] FIG. 5 is a perspective view of a portion of the nut plate assembly including a bias member, a shell, and a nut;

[0016] FIG. 6 is a perspective view of the shell;

[0017] FIG. 7 is a rear perspective view of the shell showing an inner cavity of the shell;

[0018] FIG. 8 is a side view of the shell;

[0019] FIG. 9 is a perspective view of a plate of the nut plate assembly;

[0020] FIG. 10 is a perspective view of the nut;

[0021] FIG. 11 is a bottom view of the nut;

[0022] FIG. 12 is cross-section view of the nut, taken along line A-A in FIG. 11;

[0023] FIG. 13 is a perspective view of the bias member;

[0024] FIG. 14 is a side view of the bias member;

[0025] FIG. 15 is a top view of the bias member;

[0026] FIG. 16 is a perspective view of a fastener assembly including the nut plate assembly and a fastener;

[0027] FIG. 17 is a sectional view of the fastener assembly showing the nut tilting to accommodate an orientation of the fastener;

[0028] FIG. 18 is a perspective view of the nut plate assembly, coupled to a mounting structure and a panel;

[0029] FIG. 19 is a rear perspective view of the mounting structure and the panel illustrating a fastener extending through the mounting structure and the panel and engaging the nut plate assembly;

[0030] FIG. 20 is a cross-sectional view of the nut plate assembly coupled to the mounting structure and the panel, and including the captive fastener;

[0031] FIG. 21 is a top view of an example nut plate assembly, illustrating example dimensions of the nut plate assembly;

[0032] FIG. 22 is a side view of the nut plate assembly;

[0033] FIG. 23 is a cross-sectional view of the nut plate assembly taken along line B-B shown in FIG. 21;

[0034] FIG. 24 is a side view of an example fastener, illustrating example dimensions of a fastener;

[0035] FIG. 25 is another side view of the fastener shown in FIG. 24; and
[0036] FIG. 26 is an end view of the fastener shown in FIG. 24, illustrating an example head of the fastener.

[0037] Unless otherwise indicated, the drawings provided herein are meant to illustrate features of embodiments of the disclosure. These features are believed to be applicable in a wide variety of systems comprising one or more embodiments of the disclosure. As such, the drawings are not meant to include all conventional features known by those of ordinary skill in the art to be required for the practice of the embodiments disclosed herein.

DETAILED DESCRIPTION

[0038] In the following specification and the claims, reference will be made to a number of terms, which shall be defined to have the following meanings.

[0039] The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

[0040] “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event occurs and instances where it does not.

[0041] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms such as “about,” “approximately,” and “substantially” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged; such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

[0042] Relative descriptors used herein such as upward, downward, left, right, up, down, length, height, width, thickness, and the like are with reference to the figures, and not meant in a limiting sense. Additionally, the illustrated embodiments can be understood as providing exemplary features of varying detail of certain embodiments, and therefore, features, components, modules, elements, and/or aspects of the illustrations can be otherwise combined, interconnected, sequenced, separated, interchanged, positioned, and/or rearranged without materially departing from the disclosed fastener assemblies. Additionally, the shapes and sizes of components are also exemplary and can be altered without materially affecting or limiting the disclosed technology.

[0043] The nut plate assemblies described herein overcome many of the problems associated with nut plate assemblies. In general, nut plate assemblies are used to fixedly connect panels to structures in a variety of applications such as, without limitation, aerospace applications, industrial applications, and building applications, where access to both sides of the nut plate assembly is limited or restricted. Among other features and benefits, the disclosed nut plate assemblies facilitate one or more of the use of captive panel fasteners having varying lengths, quick and easy installation and/or removal of panel fasteners, and/or single end access for blind fastening applications. In addition, nut plate assemblies described herein can accommodate fasteners that are misaligned with a center axis of the nut plate assemblies. Also, nut plate assemblies described herein are simpler to use and provide a secure connection. The advantages, and other features of the technology disclosed herein, will become more readily apparent to those having ordinary skill in the art from the following detailed description of certain preferred embodiments taken in conjunction with the drawings.

[0044] FIGS. 1-4 illustrate an example nut plate assembly. In the example, the nut plate assembly **100** includes a plate **102**, a shell **104**, a floating nut, broadly a nut, **106**, and a bias member **108**. The plate **102** includes a wall **110**. The wall **110** defines an aperture **114** for receiving a fastener (e.g., fastener **164** shown in FIGS. 16-20).

[0045] Referring to FIGS. 4 and 5, in the example, the nut plate assembly **100** is configured to

retain both the floating nut **106** and the bias member **108** within the shell **104**, which allows for the nut plate assembly **100** to be used as an inseparable assembly. In addition, during use, the nut plate assembly **100** requires no access from the nut side of nut plate assembly **100**, which is advantageous for use with panels and other structure where access to both sides of the nut plate assembly **100** is limited.

[0046] With reference to FIGS. 1-5, the shell **104** includes a first end **116**, a second end **118**, and a sidewall **120**. The sidewall **120** defines a cavity **122** and an opening **124** at the first end **116**. The shell **104** is attached to the plate **102** at the first end **116** such that the sidewall **120** extends outward from the plate **102**. As seen in FIG. 2, the plate **102** secures the floating nut **106** within the shell **104**. In the example, the shell **104** is not permanently coupled to the plate **102** and the shell **104** may be removable from the plate **102**. For example, the shell **104** includes flanges **126** that are engaged with retention tabs **128** on the plate **102**. In other examples, the shell **104** and/or the plate **102** includes retention tabs or other coupling mechanisms to secure the shell **104** to the plate **102**. For example, in some embodiments, the nut plate assembly **100** includes, without limitation, a spring, a hinge, a fastener, a clamp, and adhesive. In another example, the nut plate assembly **100** is integrally formed with or permanently affixed to the plate **102** such that the shell **104** and the plate **102** cannot be separated without damaging the nut plate assembly **100**.

[0047] As illustrated in FIGS. 5-9, the sidewall **120** of the shell **104** is shaped to enclose and retain the floating nut **106** on the plate **102**. For example, at the first end **116**, the sidewall **120** is perpendicular to a plane of the plate **102**. At the second end **118**, the sidewall **120** tapers radially inward, e.g., by a swaging process, to a closed end to facilitate retaining the floating nut **106** within the shell **104** when the shell **104** is coupled to the plate **102**. In addition, the shell **104** is shaped to engage and prevent at least some rotation of the floating nut **106**. For example, the sidewall **120** includes shoulders **130** that project outward from the cylindrical main body of the shell **104** and define side cavities arranged to receive shoulders **132** on the floating nut **106**. The floating nut **106** engages the shoulders **130** of the shell **104** and is fixed against rotation relative to the shell **104**.

[0048] Also, in the example, the floating nut **106** is disposed within the shell **104** and is moveable relative to the sidewall **120**. For example, the floating nut **106** is moveable along the center axis **134** of the shell **104** between a first position and a second position. In the first position, the floating nut **106** is proximate the second end **118**. In the second position, the floating nut **106** is proximate the first end **116**. In another example, the floating nut **106** is moveable between a first position proximate the first end **116** and a second position proximate the second end **118**. In addition, in the illustrated example, the floating nut **106** is supported by the bias member **108** at the first end **116** of the shell **104** and the floating nut **106** is tiltable about the center axis **134**. The shell **104** inhibits rotation of the floating nut **106** about the center axis **134**. Accordingly, in the example, the floating nut **106** is free to move in at least one direction when the floating nut **106** is secured within the cavity **122** of the shell **104**. For example, the floating nut **106** is free to move axially within the shell **104** and tilt at an angle relative to the center axis **134** but is inhibited from rotating about the center axis **134**, as shown in FIG. 17. The freedom of movement of the floating nut **106** facilitates the floating nut **106** accommodating misalignment of a fastener (e.g., fastener **164** shown in FIGS. 16-20) and the nut plate assembly **100** and/or joining panels with surfaces that are not parallel. The restriction of rotation of the floating nut **106** facilitates the floating nut **106** engaging a fastener **164** and enabling quick and simple securement of the fastener **164** to the nut plate assembly **100**.

[0049] In the example, the bias member **108** is positioned within the shell **104** and between the floating nut **106** and the first end **116** of the shell **104**. As such, the bias member **108** biases the floating nut **106** axially away from the plate **102** and towards the second end **118**. During use of the nut plate assembly **100**, the bias member **108** facilitates the floating nut **106** engaging the fastener **164** (shown in FIG. 18). In the example, the bias member **108** is shaped to provide a desired pre-load force on the floating nut **106**, as shown in FIG. 13. Alternatively, the bias member **108** may be any type of bias or force provider that enables nut plate assembly **100** to function as described

herein. For example, the bias member **108** may be a compression spring. In some examples, the pre-load force on the floating nut **106** may be adjusted by varying the wire diameter and spring length of the bias member **108**. In the example, the characteristics and shape of bias member **108** are selected to provide the necessary pre-load force required for operation of nut plate assembly **100**, while maintaining the bias member **108** in an axial resiliency range.

[0050] With reference, to FIGS. **10-12**, the floating nut **106** has a body **140** that defines an inner bore **142** therethrough. The body **140** includes a first end **144** and a second end **146**. The bore **142** extends between and through the first end **144** and the second end **146** of the body **140**. At the first end **144** of the body **140**, the bore **142** includes a tapered portion configured to facilitate receiving and aligning a fastener **164** (shown in FIGS. **18-20**) with the body **140**. The body **140** includes an inner surface **141** defining the bore **142** and shaped to receive the fastener **164**. In addition, the inner surface **141** defines at least one groove **143** extending along the bore and parallel to the center axis **134**. In the example, the floating nut **106** includes a pair of the grooves **143** that are spaced circumferentially apart by 180°.

[0051] Referring to FIGS. **10-12** and **17**, in the example, the floating nut **106** includes two shoulders **132** extending from the second end **146** axially along the center axis **134** a predetermined distance. The shoulders **132** are sized to be received within shoulders **130** of the shell **104** (shown in FIGS. **6** and **7**) and engage the shell **104**. The shoulders **130** of the shell **104** are longer than the shoulders **132** of the floating nut **106** to facilitate axial movement of the floating nut **106** within the shell **104**. The shoulders **130** of the shell **104** and the shoulders **132** are sized and shaped to restrict rotation of the floating nut **106** within the shell **104**. For example, the shoulders **132** of the floating nut **106** have a shape that matches the shape of the shoulders **130** of the shell **104** and are only slightly smaller to provide a close fit between the shoulders **132** and the shoulders **130** of the shell **104**. Accordingly, the inner surfaces of the shoulders **130** of the shell **104** contact the shoulders **132** of the floating nut **106** and prevent rotation of the floating nut **106** relative to the shell **104**.

[0052] As shown in FIGS. **5-7**, the shoulders **132** of the floating nut **106** define recesses and include bearing surfaces **160** that are shaped to receive the bias member **108**. In the example, the bearing surfaces **160** are sized to prevent over-stressing and/or damaging the bias member **108** when the bias member **108** is compressed, while enabling the bias member **108** to urge the body **140** away from the plate **102** when extended. The shoulders **132** and the bearing surfaces **160** enable the bias member **108** to apply an axial force to the body **140** to urge the body **140** away from the plate **102**.

[0053] In one example, the floating nut **106** is a single piece. In another example, the floating nut **106** includes at least two pieces. For example, the bore **142** and/or the bearing surfaces **160** may be defined by an insert, a washer, and/or another member that is coupled to body **140** of the floating nut **106**.

[0054] As seen in FIGS. **5** and **11**, in the example, the sidewall **120** of the shell **104** defines a cross-sectional shape of the shell **104** and is configured to engage the floating nut **106**. The shell **104** and the floating nut **106** define a gap therebetween that is sized to allow axial movement of the floating nut **106** and tilting of the floating nut **106** relative to the shell **104**. The shoulders **132** of the floating nut **106** are received within the shoulders of the shell **104** and prevent rotation of the floating nut **106** relative to the shell **104**. Accordingly, the sidewall **120** of the shell **104** engages the floating nut **106** within the shoulders to inhibit rotation of the floating nut **106** when the floating nut **106** is positioned within the shell **104**. In the example, the cross-sectional shape of the shell **104** is a cylinder with the shoulders extending outward from the cylinder. The shell **104** may be other shapes without departing from some aspects of the disclosure. For example, in some examples, the shell **104** includes, without limitation, a cylinder, a triangle, a rectangle, a trapezoid, a pentagon, a hexagon, a heptagon, an octagon, a nonagon, a decagon, a dodecagon, and a star.

[0055] Referring to FIGS. **5** and **13-15**, in the example, the bias member **108** is formed of a flexible and resilient wire. The bias member **108** is sized and shaped to be received within the shell **104** and

engage the floating nut **106** to provide a desired bias force to the floating nut **106**. For example, the bias member **108** comprises an open spring structure including a planar portion forming a main body **150** and angled portions forming wings **152** extending outward from the main body **150**. The main body **150** is in a plane and the wings **152** extend at an angle relative to the plane of the main body **150**. In example, the bias member **108** is an open structure in which the middle portions of the main body **150** and the wings **152** are open. In other embodiments, the bias member **108** may include one or more flexible plates or other structures that cover at least some of the structure and/or form a part of the planar or angled portions.

[0056] In the illustrated example, the bias member **108** is disposed between the plate **102** (shown in FIG. **1**) and the floating nut **106**, and is configured to bias the floating nut **106** towards the second end **118** of the shell **104**. For example, the main body **150** engages the bearing surfaces **160** of the floating nut **106** and the wings **152** engage the plate **102**. The wings **152** are compressed or bent when an axial force is applied to the floating nut **106** to facilitate the floating nut **106** moving toward the plate **102**. The bias member **108** is resilient and returns to its original shape and maintains a predetermined distance between the floating nut **106** and the plate **102** when the axial force is removed.

[0057] With reference to FIGS. **16** and **17**, the nut plate assembly **100** is arranged to receive and engage the fastener **164**. For example, the fastener **164** extends through the aperture in the plate **102** and engages the floating nut **106**. The floating nut **106** is able to tilt relative to the center axis **134** to accommodate misalignment of the fastener **164** relative to the center axis **134** of the nut plate assembly **100**. For example, the floating nut **106** can tilt up to 10° relative to the center axis **134** to accommodate angular misalignment. In addition, the nut plate assembly **100** can also accommodate axial misalignment.

[0058] Referring to FIGS. **16-20**, the fastener **164** includes a head **154**, a shank **156** extending axially from the head **154**, and at least one protuberance **159** extending outward from the shank **156** at an axial distance from the head **154**. The fastener **164** is rotatable relative to the floating nut **106** between a locked orientation and an unlocked orientation. In the example, the fastener **164** is rotatable 90° (a quarter-turn) between the locked orientation and the unlocked orientation. As result, the fastener **164** (and nut plate assembly **100** shown in FIG. **1**) provide a locking arrangement that facilitates quickly and simply securing components using the fastener assembly. For example, fastener **164** needs to be rotated less than a full turn (e.g . . . a quarter-turn) to move into the locked position. In addition, an installer may recognize that the fastener **164** is in a fully locked position when the head **154** of the fastener **164** is turned. In contrast, conventional fasteners must be repeatedly rotated multiple turns to achieve a desired tightness and it is unclear if the Examiner has fully locked the fastener after multiple turns.

[0059] In the unlocked orientation, the shank **156** of the fastener **164** is able to move axially through the bore **142** of the floating nut **106**. For example, the protuberances **159** extend through the axial grooves **143** along the inner surface **141** of the floating nut **106**. The grooves **143** are sized to receive the protuberances. For example, the grooves **143** each have a width that is larger than a width of the protuberances **159**.

[0060] The floating nut **106** includes at least one engagement surface **170** that contacts and engages the protuberances **159** of the fastener **164** when the fastener **164** is in the locked orientation. In the example, the floating nut **106** includes a pair of engagement surfaces **170**. For example, each engagement surface **170** is an annular surface or ledge that extends radially outward from the inner surface **141** of the floating nut **106** and circumferentially between the grooves **143**. The engagement surfaces **170** extend at an angle relative to the inner surface **141** of the floating nut **106**. For example, the engagement surfaces **170** are perpendicular to the inner surface **141** of the floating nut **106**. In the example, the engagement surfaces **170** form a ledge adjacent to the inner surface **141** of the floating nut **106**. The wall **110** extends beyond the ledge formed by the engagement surfaces **170**. The engagement surfaces **170** each engage a respective one of the

protuberances **159** and prevent the fastener from moving axially relative to the floating nut **106**. [0061] FIGS. **18** and **19** are perspective views of the nut plate assembly **100** installed and used to couple a panel **168** to a mounting structure **162** via the captive fastener **164**. FIG. **20** is a cross-sectional view of the nut plate assembly **100** secured to the mounting structure **162** and the panel **168**. In the example, the plate **102** of the nut plate assembly **100** is mechanically coupled to the mounting structure **162** by fasteners such as, for example, and without limitation, nut and bolt combinations, sheet metal fasteners, rivets, and the like. In another example, the nut plate assembly **100** is coupled to the mounting structure **162** by, for example, and without limitation, adhesive bonding or any other suitable bonding materials.

[0062] The mounting structure **162** includes an aperture **166** defined therethrough and sized to receive at least a portion of the fastener **164**. The fastener **164** also extends through the aperture **166** of the panel **168**. The nut plate assembly **100** is secured to the mounting structure **162** such that the aperture **114** of the plate **102** is aligned with the aperture **166** of the mounting structure **162** and the nut plate assembly **100** is arranged to receive the fastener **164** protruding from the aperture **166**. In some examples, the nut plate assembly **100** includes, for example, and without limitation, a retaining ring, an E-clip, a spring plunger, and/or any mechanism configured to facilitate capturing the fastener **164** in the panel **168**.

[0063] In the example, the fastener **164** is a panel bolt having the head **154** as shown in FIG. **16**. Alternatively, the fastener **164** is any type of fastener having the head **154** taking any shape or form, including for example, and without limitation, a hexagonal head, a spline head, a flat head, a socket cap head, and a pan head or a headless fastener. In some examples, the fastener **164** is a locking fastener, including one or more components configured to lock fastener **164** against rotation relative to panel **168**.

[0064] The panel **168**, with the fastener **164**, is introduced to the mounting structure **162** with the nut plate assembly **100** secured on the mounting structure **162**. The fastener **164** is aligned with the floating nut **106** and the panel **168** and introduced into the nut plate assembly **100**. The bias member **108** is arranged to bias the floating nut **106** toward the mounting structure **162** to provide a preload of the floating nut **106**. In addition, the floating nut **106** can tilt or move radially relative to the center axis **134** to accommodate if the fastener **164** is positioned at angle or off-axis relative to the center axis **134**. The fastener **164** is pushed toward the mounting structure **162** until the fastener **164** moves through the bore **142** of the floating nut **106** and the protuberances of the fastener **164** move through and beyond the grooves **143** of the floating nut **106**. The fastener **164** is rotated from the unlocked orientation to the locked orientation and the protuberances **159** engage the engagement surfaces **170** of the floating nut **106** to secure the panel **168** and the mounting structure **162** together. In the locked orientation, the engagement surfaces **170** engage the protuberances **159** and prevent the fastener **164** from moving axially relative to the floating nut **106**. In some examples, the engagement surfaces **170** are shaped to resist rotation of the fastener **164** and maintain the fastener **164** in the locked orientation until a suitable force is delivered to the fastener **164** and/or the floating nut **106**. For example, in the illustrated embodiment, due to the position of the bias member, the engagement surfaces **170** are biased toward the protuberances **159** and provide a force on the fastener **164** to prevent movement of the fastener **164** relative to the floating nut **106**.

[0065] To release the fastener **164**, the fastener **164** is rotated from the unlocked orientation to the unlocked orientation. In the unlocked orientation, the protuberances **159** of the fastener **164** are able to move axially through the grooves and the fastener **164** may be removed from the floating nut **106**.

[0066] In the example, the fastener **164** may rotate in a clockwise or counterclockwise direction to move between the unlocked orientation and the locked orientation. In other examples, the fastener **164** is rotated in one direction to move from the unlocked orientation to the locked orientation and rotated in the opposite direction to move from the locked orientation to the unlocked orientation. In

further examples, the fastener **164** is rotated in a single direction to move between the unlocked orientation and the locked orientation.

[0067] To assemble the nut plate assembly **100**, the bias member **108** and the floating nut **106** are placed into the shell **104** and the plate **102** is coupled to the shell **104** to capture the bias member **108** and the floating nut **106**. The bias member **108** engages the plate **102** and the floating nut **106** and is compressed within the shell **104**. The opening **124** of the shell **104** is aligned with the aperture **114** of the plate **102**, and arranged to receive the fastener **164**. In the example, the aperture **114** is oversized in comparison to the fastener **164** (i.e., the aperture **114** is larger than a size necessary to accommodate the diameter of the fastener). Accordingly, the aperture **114** facilitates accommodating increased off-axis engagement of the fastener **164** in the nut plate assembly **100**.

[0068] The fastener **164** may be captured in the nut plate assembly **100** to form a fastener assembly **200**. For example, the fastener **164** is positioned through the aperture **114** of the plate **102**, the opening **124** of the shell **104**, and into the bore **142** of the floating nut **106**. The fastener **164** is moved through bore **142** in the unlocked orientation such that the protuberance **159** of the fastener **164** is moved through the groove **143** until the protuberance **159** is beyond the groove **143**. The fastener **164** is then rotated from the unlocked orientation to the locked orientation and the protuberance(s) **159** of the fastener **164** engages the engagement surfaces **170** of the floating nut **106**. The bias member **108** biases the floating nut **106** toward the second end **118** of the shell **104** to facilitate engagement of the floating nut **106** and the fastener **164** and secure the fastener **164** in position. For example, the floating nut **106** is biased such that the engagement surfaces **170** of the floating nut **106** are pressed against the protuberance **159** of the fastener **164**. The engagement surfaces **170** may be shaped to inhibit rotation of the fastener **164** and prevent the fastener **164** from moving to the unlocked orientation unless a predetermined force is provided to the fastener **164**.

[0069] FIGS. 21-23 illustrates example dimensions of a nut plate

[0070] assembly (e.g., nut plate assembly **100**). FIG. 23 is a cross-sectional view of the nut plate assembly taken along line A-A in FIG. 21. For example, the nut plate assembly **100** has an overall height **H1** measured from a bottom surface of the plate **102** to the second end **118** of the shell **104**. The shell **104** has a diameter **D1** at the second end **118** of the shell **104** and a height **H2** measured from the first end **116** to the second end **118** of the shell **104**. Also, the shell **104** has a width **W1** at the shoulders **132** that is larger than the diameter **D1** at the second end **118** of the shell **104**. The shoulders **132** have a height **H3** that is less than the height **H1** of the shell **104**. Also, the shoulders **132** each have a width **W2** that is less than the diameter **D1** of the shell **104**. Accordingly, the shoulders **132** are shaped to engage the floating nut **106** and prevent rotation of the floating nut **106** while facilitating axial movement and tilting of the floating nut **106**.

[0071] In addition, the plate **102** is shaped to be secured to or otherwise affixed to a substrate and to receive the shell **104** and floating nut **106**. For example, the plate **102** includes openings **172** spaced apart a distance **D2** and positioned on opposite sides of the shell **104** when the shell **104** is coupled to the plate **102**. The openings **172** have a diameter **D3** that is sized to receive fasteners or other securement devices. In addition, sides of the plate **102** adjacent the shoulders **132** of the shell **104** may be curved with a radius **R1** that matches a radius of at least one curve on the shell **104**. The sides of the plate **102** adjacent the openings **172** may be curved with a radius **R2** that is smaller than the radius **R1** such that the plate **102** projects outward and accommodates the openings **172**. Also, the plate **102** has a thickness **T1** that facilitates the plate **102** supporting the floating nut **106** and the shell **104**. In other embodiments, the plate **102** may have any suitable shape. For example, in some embodiments, the openings **172** are omitted and the plate **102** is bonded to the structure or panel.

[0072] FIGS. 24-26 illustrates example dimensions and shapes of a fastener (e.g., fastener **164**). The fastener **164** is sized to extend through and engage a nut plate assembly (e.g., the nut plate assembly **100** shown in FIGS. 21-23). For example, the fastener **164** has a length **L1** measured

from the head **154** to an end of the fastener **164**. The head **154** has a diameter **D4**, and the shank **156** has a diameter **D5**.

[0073] The protuberances **159** extend radially outward from the shank **156** and are sized to engage the floating nut **106** (shown in FIGS. **21-23**). For example, the protuberances have an axial length **L2** and a width **W3**. In addition, the fastener **164** has width **W4** at the protuberances **159** that is less than the diameter **D4** of the head **154**. The fastener **164** is sized to fit through a groove in a nut plate assembly in an unlocked orientation and engage engagement surfaces in a locked orientation.

[0074] In a first example, a nut plate assembly comprises a plate comprising an aperture defined therethrough; a shell attached to the plate and extending along a center axis; and a nut retained within the shell and comprising an inner surface and an engagement surface that extends at an angle relative to the inner surface, the inner surface defining an inner bore and having a groove extending along the inner bore and parallel to the center axis, wherein the nut plate assembly is configured to receive a fastener extending through the aperture of the plate and into the inner bore in the nut, wherein in an unlocked orientation, the fastener is configured to move through the groove, and wherein in a locked orientation, the engagement surface engages the fastener and prevents the fastener moving axially relative to the floating nut.

[0075] In a second example, the fastener is rotated within the inner bore between the locked orientation and the unlocked orientation.

[0076] In a third example, the unlocked orientation and the locked orientation are 90° apart.

[0077] In a fourth example, the groove is sized to receive a protuberance on the fastener, and wherein the engagement surface is configured to engage the protuberance when the nut plate assembly is in the locked orientation.

[0078] In a fifth example, a bias member is disposed within the shell and engaged with the nut, wherein the bias member is disposed between the plate and the nut and is configured to bias the nut towards an end of the shell.

[0079] In a sixth example, the nut is tiltable within the shell and relative to the center axis to engage the fastener if the fastener extends at an angle relative to the center axis.

[0080] In a seventh example, the nut is able to move axially within the shell and is fixed against rotation relative to the shell.

[0081] In an eighth example, the nut includes a first end, a second end, and a shoulder, the shoulder is shaped to engage the shell and fix the nut against rotation within the shell.

[0082] In a ninth example, a fastener assembly comprises a plate comprising an aperture defined therethrough; a shell attached to the plate and extending along a center axis; a nut retained within the shell and comprising an inner bore; a fastener extending through the plate and into the shell to engage the nut; and a bias member disposed within the shell and engaged with the nut, wherein the fastener is rotatable between an unlocked orientation and a locked orientation when the fastener is engaged with the nut, the nut includes an engagement surface that engages the fastener when the fastener is in the locked orientation, wherein the unlocked orientation and the locked orientation are 90° apart.

[0083] In a tenth example, the nut includes an inner surface defining the inner bore, the inner surface having a groove extending along the inner bore and parallel to the center axis.

[0084] In an eleventh example, the fastener includes a protuberance sized to move through the groove, wherein the protuberance engages the engagement surface of the nut when the fastener is in the locked orientation.

[0085] In a twelfth example, the bias member is disposed between the plate and the nut, and is configured to bias the nut towards an end of the shell.

[0086] In a thirteenth example, the nut is tiltable within the shell and relative to the center axis to engage the fastener if the fastener extends at an angle relative to the center axis.

[0087] In a fourteenth example, the nut includes a first end, a second end, and a shoulder, the shoulder is shaped to engage the shell and fix the nut against rotation within the shell.

[0088] In a fifteenth example, the first end of the nut has a recess arranged to receive the bias member.

[0089] In a sixteenth example, the bias member comprises a wire spring formed of a flexible and resilient wire.

[0090] In a seventeenth example, the bias member includes a main body and wings extending outward from the main body.

[0091] In an eighteenth example, a method of assembling a nut plate assembly comprises positioning a nut within a shell extending along a center axis, the nut comprising an inner surface and an engagement surface that extends at an angle relative to the inner surface, the inner surface defining an inner bore and having a groove extending along the inner bore and parallel to the center axis; and coupling a plate to the shell to capture the nut within the shell, the plate comprising an aperture arranged to receive a fastener such that the fastener extends through the plate and into the shell to engage the nut, wherein in an unlocked orientation, the fastener is configured to move through the groove, and wherein in a locked orientation, the engagement surface engages the fastener and prevents the fastener moving axially relative to the floating nut.

[0092] In a nineteenth example, a method includes positioning a bias member within a shell, the nut configured to engage the bias member, wherein the bias member is arranged to bias the nut toward an end of the shell, and wherein the nut is tiltable within the shell and relative to the center axis to engage the fastener if the fastener extends at an angle relative to the center axis.

[0093] In a twentieth example, a method includes fixing the nut against rotation within the shell, wherein the fastener is rotatable relative to the nut between an unlocked orientation and a locked orientation when the fastener is engaged with the nut, the nut includes an engagement surface that engages the fastener when the fastener is in the locked orientation.

[0094] The components as described herein provide spring-loaded nut plate assemblies. For example, as described in the embodiments herein, a floating nut of the nut plate assemblies is biased by a bias member, which enables the use of captive panel screws. This facilitates ease of assembly and disassembly of a panel to an underlying structure. In addition, the spring-loaded nut plate assemblies facilitate the nut plate assembly accommodating varying lengths of captive panel fasteners. The bias member facilitates one of pulling the structure components together or pushing them apart during assembly or removal of the panel structure.

[0095] The nut plate assemblies facilitate engagement with fasteners extending at an angle relative to a center axis or axially offset from the center axis. In addition, the nut plate assemblies provide a secure and simple engagement of the fastener and the nut plate assemblies. Moreover, the nut plate assemblies and fasteners do not require threads or complicated components that may be damaged due to improper handling or installation. As a result, the nut plate assemblies provide a reliable manner to secure components together and reduces time and effort required for assembly.

[0096] Examples of spring-loaded nut plate assemblies are described above. The systems and methods are not limited to the specific embodiments described herein, but rather, components of the systems and/or operations of the methods may be utilized independently and separately from other components and/or operations described herein. Further, the described components and/or operations may also be defined in, or used in combination with, other systems, methods, and/or devices, and are not limited to practice with only the systems described herein.

[0097] Although specific features of various embodiments of the disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the disclosure, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

[0098] This written description uses examples to disclose the embodiments, including the best mode, and also to enable any person skilled in the art to practice the embodiments, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that

occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A nut plate assembly comprising: a plate comprising an aperture defined therethrough; a shell attached to the plate and extending along a center axis; and a nut retained within the shell and comprising an inner surface and an engagement surface that extends at an angle relative to the inner surface, the inner surface defining an inner bore and having a groove extending along the inner bore and parallel to the center axis, wherein the nut plate assembly is configured to receive a fastener extending through the aperture of the plate and into the inner bore in the nut, wherein in an unlocked orientation, the fastener is configured to move through the groove, and wherein in a locked orientation, the engagement surface of the nut engages the fastener and prevents the fastener moving axially relative to the nut.
2. A nut plate assembly in accordance with claim 1, wherein the fastener is rotated within the inner bore between the locked orientation and the unlocked orientation.
3. A nut plate assembly in accordance with claim 2, wherein the unlocked orientation and the locked orientation are 90° apart.
4. A nut plate assembly in accordance with claim 1, wherein the groove is sized to receive a protuberance on the fastener, and wherein the engagement surface is configured to engage the protuberance when the nut plate assembly is in the locked orientation.
5. A nut plate assembly in accordance with claim 1, further comprising a bias member disposed within the shell and engaged with the nut, wherein the bias member is disposed between the plate and the nut and is configured to bias the nut towards an end of the shell.
6. A nut plate assembly in accordance with claim 5, wherein the nut is tiltable within the shell and relative to the center axis to engage the fastener if the fastener extends at an angle relative to the center axis.
7. A nut plate assembly in accordance with claim 1, wherein the nut is able to move axially within the shell and is fixed against rotation relative to the shell.
8. A nut plate assembly in accordance with claim 7, wherein the nut includes a first end, a second end, and a shoulder, the shoulder is shaped to engage the shell and fix the nut against rotation within the shell.
9. A fastener assembly comprising: a plate comprising an aperture defined therethrough; a shell attached to the plate and extending along a center axis; a nut retained within the shell and comprising an inner bore; a fastener extending through the plate and into the shell to engage the nut; and a bias member disposed within the shell and engaged with the nut, wherein the fastener is rotatable between an unlocked orientation and a locked orientation when the fastener is engaged with the nut, the nut includes an engagement surface that engages the fastener when the fastener is in the locked orientation, wherein the unlocked orientation and the locked orientation are 90° apart.
10. A fastener assembly in accordance with claim 9, wherein the nut includes an inner surface defining the inner bore, the inner surface having a groove extending along the inner bore and parallel to the center axis.
11. A fastener assembly in accordance with claim 10, wherein the fastener includes a protuberance sized to move through the groove, wherein the protuberance engages the engagement surface of the nut when the fastener is in the locked orientation.
12. A fastener assembly in accordance with claim 9, wherein the bias member is disposed between the plate and the nut, and is configured to bias the nut towards an end of the shell.
13. A fastener assembly in accordance with claim 9, wherein the nut is tiltable within the shell and

relative to the center axis to engage the fastener if the fastener extends at an angle relative to the center axis.

14. A fastener assembly in accordance with claim 9, wherein the nut includes a first end, a second end, and a shoulder, the shoulder is shaped to engage the shell and fix the nut against rotation within the shell.

15. A fastener assembly in accordance with claim 14, wherein the first end of the nut has a recess arranged to receive the bias member.

16. A fastener assembly in accordance with claim 9, wherein the bias member comprises a wire spring formed of a flexible and resilient wire.

17. A fastener assembly in accordance with claim 9, wherein the bias member includes a main body and wings extending outward from the main body.

18. A method of assembling a nut plate assembly, said method comprising: positioning a nut within a shell extending along a center axis, the nut comprising an inner surface and an engagement surface that extends at an angle relative to the inner surface, the inner surface defining an inner bore and having a groove extending along the inner bore and parallel to the center axis; and coupling a plate to the shell to capture the nut within the shell, the plate comprising an aperture arranged to receive a fastener such that the fastener extends through the plate and into the shell to engage the nut, wherein in an unlocked orientation, the fastener is configured to move through the groove, and wherein in a locked orientation, the engagement surface engages the fastener and prevents the fastener moving axially relative to the nut.

19. A method in accordance with claim 18, further comprising positioning a bias member within a shell, the nut configured to engage the bias member, wherein the bias member is arranged to bias the nut toward an end of the shell, and wherein the nut is tiltable within the shell and relative to the center axis to engage the fastener if the fastener extends at an angle relative to the center axis.

20. A method in accordance with claim 18, further comprising fixing the nut against rotation within the shell, wherein the fastener is rotatable relative to the nut between an unlocked orientation and a locked orientation when the fastener is engaged with the nut, the nut includes an engagement surface that engages the fastener when the fastener is in the locked orientation.
